protons, the ordinates show the difference between neutrons and protons, $N - Z$. Stable nuclei are indicated by black dots, electron emitters by upright triangles (the apex pointing upward), and positron emitters by inverted triangles. Since the electron emitters are relatively rich in neutrons, whereas the positron emitters contain relatively few, the former appear mostly in the region where $N - Z$ is larger—i.e. on the upper periphery of the group of nuclei—while the majority of the positron emitters are found at the lower edge of the group, where $N - Z$ is smaller. It is a striking fact that not infrequently unstable nuclei are also found interspersed among the stable ones. The reasons for this circumstance will be discussed later. Certain nuclei change by capturing an electron from the innermost electron shell (the *K-shell*) of their extranuclear structure. These nuclei are referred to as *K-capturers,* and are indicated in our tables by circles. We find also a number of squares, mostly at the end of the table, indicating nuclei which emit alpha rays.

Up to this point, we have discussed nuclear stability in its relation to transmutation by the emission of electrons or positrons. Let us now briefly discuss stability in its relation to transmutations by alpha ray emission. Studying Figure 11, we can conclude from the initial decrease and subsequent increase in the magnitude of the term $E/(N + Z)$ that nuclear stability first increases steadily, up to about where $Z + N = 40$, but thereafter it gradually decreases, because of the increasing effect of electrical repulsion. Nevertheless, even in these cases, work has to be done in order to remove an individual particle from the nucleus. Let us assume that after applying the necessary work, we remove simultaneously two neutrons and two protons from the nucleus and then combine them in a helium nucleus. In this process a very large quantity of energy, 30 Mev., is liberated. If this energy is greater than that quantity of energy required in order to separate four individual particles, the net result is an actual gain in energy. Therefore, from the point of view of energetics, this process is an advantageous one and ought to be taking place spontaneously, in the form of the formation of an alpha particle in the nucleus and its subsequent emission by the latter. The probability of the occurrence of this process is bound to increase with the increase of the number of particles, for the binding energy per particle decreases as the total number of particles increases. Therefore, alpha emitters are to be looked for among the heavier nuclei, in conformity with actual experience. Actually, as we approach the heaviest nuclei, the magnitude of binding energy per particle drops—approximately at least—to a value between 6 and 7 Mev., in other words to about one-fourth of the binding energy of a helium nucleus.

When the total number of particles is high, the splitting of a nucleus into two parts, not very different in size, is advantageous from the point of view of energetics. A nucleus of mass number 230 might split into two nuclei, one of